

TFL London Underground “What If?” Simulator

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Project description

Maintaining and communicating the efficiency of the London Underground requires tools that can rapidly model the impact of unexpected closures alongside real-time transit conditions.

This project is a web-based interactive visualiser and simulator using Next.js, Neo4j and Leaflet. The system integrates live Transport for London (TfL) feeds to display real-time train movements across the network. It features a dedicated "What-If" mode, enabling users to sever graph connections by simulating targeted station or line closures.

The expected outcome is a highly responsive application that recalculates shortest-path routes via Dijkstra’s algorithm, dynamically avoiding user-defined disruptions. By bridging live tracking with interactive scenario planning, the project should deliver a comprehensive tool for analysing transit resilience, improving journey predictability and providing immediate visual feedback on network health.